

Recovering critical materials with a modular approach

SPHERE - METALS AND MINERALS

Nordic Salt Cycle is developing a molten-salt technology that recovers valuable metals from battery black mass and other complex industrial waste streams.

By focusing on the refining step and integrating with existing infrastructure, the company aims to make critical-material recovery more efficient, flexible and capital-light.

90%+
OVERALL METAL RECOVERY

99.5%+
LITHIUM CARBONATE PURITY

~€4M
CAPITAL RAISED TO DATE

PROVEN POTENTIAL

Unlocking Value in the Electrification Transition

Europe needs reliable supplies of lithium, nickel and cobalt for batteries and other strategic industries.
Nordic Salt Cycle develops and operates modular molten-salt recovery plants, co-located with feedstock suppliers. By processing materials close to their source, the model avoids costly feedstock transport and makes recovery viable at a smaller scale than conventional battery recycling technologies.

1000 t/year

CAPACITY OF EACH MODULE

5000 t/year

EXPECTED CAPACITY BY 2030
PER SITE



A modular, decentralised model designed to recover critical materials closer to the source, reducing logistics costs and enabling smaller-scale operations.

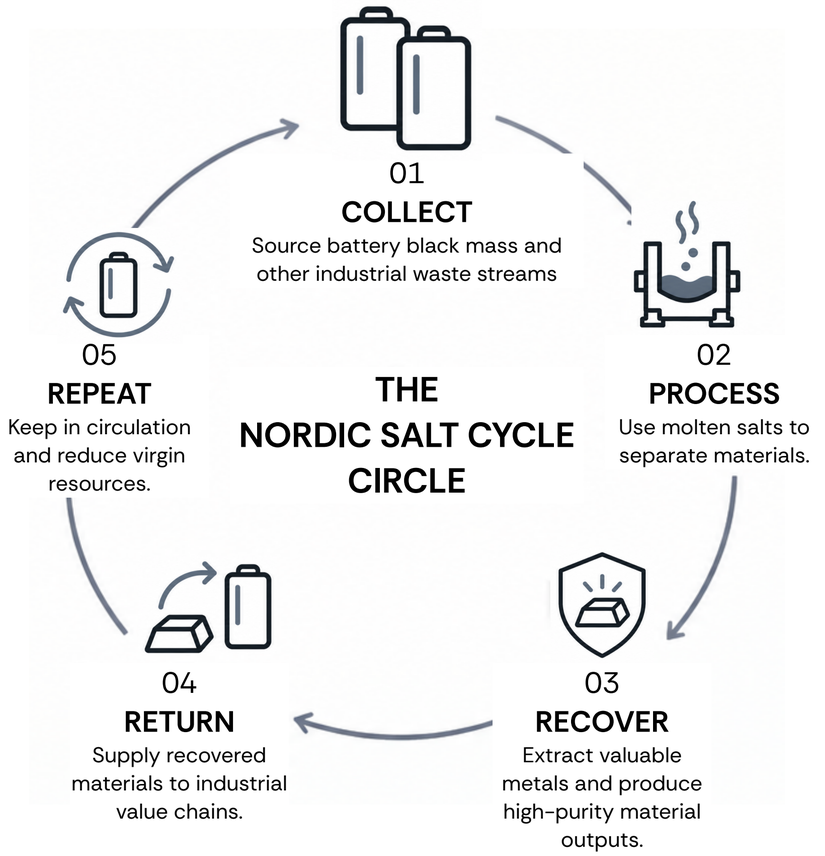
THE CIRCULAR MODEL

One Platform. Multiple Waste Streams

Nordic Salt Cycle's molten-salt technology has been developed primarily for battery recycling, with demonstrated potential to recover critical materials from other complex industrial waste streams.

Beyond batteries, the company is already working with a major industrial player to recover tungsten and address specific material recovery challenges.

This broader applicability positions Nordic Salt Cycle as a platform technology, with potential applications beyond the battery recycling market.



From laboratory results to repeatable industrial modules

Can Nordic Salt Cycle turn successful technical trials into reliable industrial operations, competitive cost per tonne and repeatable modular plants?



2024
FOUNDED

Nordic Salt Cycle starts developing its molten-salt process for critical-material recovery.

2024-26
EARLY FINANCING

€4M
SEED CAPITAL

Backed by EIFO, Ananda Impact Ventures and The Footprint Firm.

PILOT
SCALE UP

88x
PROTOTYPING

The main reactor has been scaled up 88× from prototype to pilot. The next step to commercial scale is designed to be a further 3.6×.

NEXT
PROVE INDUSTRIAL ECONOMICS

European refiners have confirmed compatibility with their processes and expressed interest in purchasing the material at scale.

2030
FIRST COMMERCIAL SITE

5,000 t
PER YEAR CAPACITY

The company expects an initial site with 5,000 tonnes per year of capacity, designed around repeatable 1,000-tonne-per-year modules.

Recovering valuable metals from waste streams keeps critical materials in circulation and can reduce dependence on virgin mining.

WHY IT WORKS

Better recovery only matters if the economics work

The differentiator is the combination of high recovery rates, modular deployment and a platform technology with potential applications across multiple industrial waste streams.

01 — RECOVERY

Higher material recovery

More than 90% overall metal recovery has been demonstrated from NMC black mass, with reported product purities above 95% for nickel-cobalt product and 99.5% for lithium carbonate.

02 — MODULAR DEPLOYMENT

Modular deployment

Repeatable 1,000-tonne-per-year modules are designed to integrate into existing industrial infrastructure, potentially reducing the need for large, fully integrated recycling plants.

03 — PLATFORM TECHNOLOGY

Beyond battery recycling

Developed primarily for battery recycling, the technology has also demonstrated the ability to recover critical materials from other industrial waste streams. Work on tungsten recovery illustrates its potential to address material recovery challenges beyond batteries.

03 — RELEVANCE

Strategic relevance

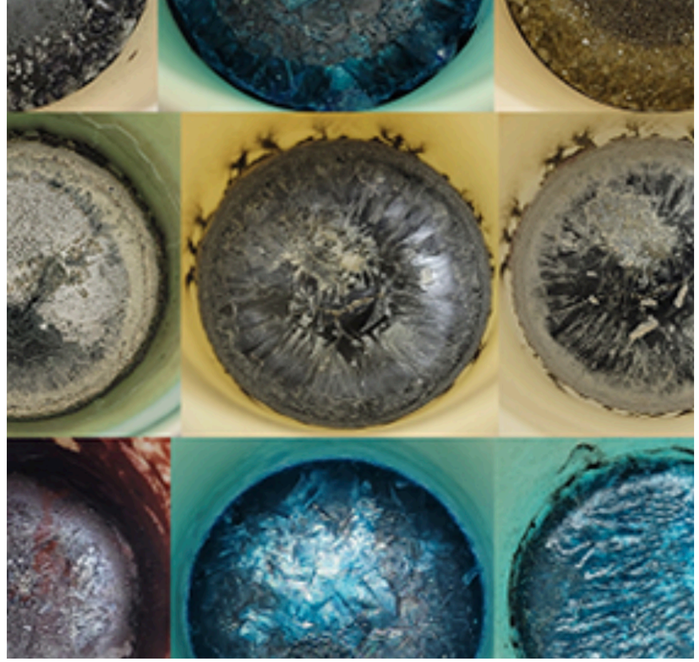
Recovering critical materials can support European supply-chain resilience and reduce dependence on imported raw materials, while creating a commercial proposition beyond sustainability alone.

STRATEGIC PARTNERSHIPS

Backing from EIFO, Ananda Impact Ventures and The Footprint Firm combines early-stage capital with experience in industrial scale-up and impact investing.

THE INVESTOR TAKEAWAY

Prove the process, then replicate the module



Nordic Salt Cycle has made meaningful technical progress. The next stage is to prove that the process can operate reliably and profitably at commercial scale.

01 LOOK FOR TECHNICAL PROOF

Strong recovery and purity results are encouraging, but the investment case depends on whether they hold under continuous industrial operation.

02 LOOK FOR MODULARITY TO BUILD SOLID ECONOMY

Adding refining capacity to existing facilities may reduce the capital and execution burden compared with building a complete recycling chain.

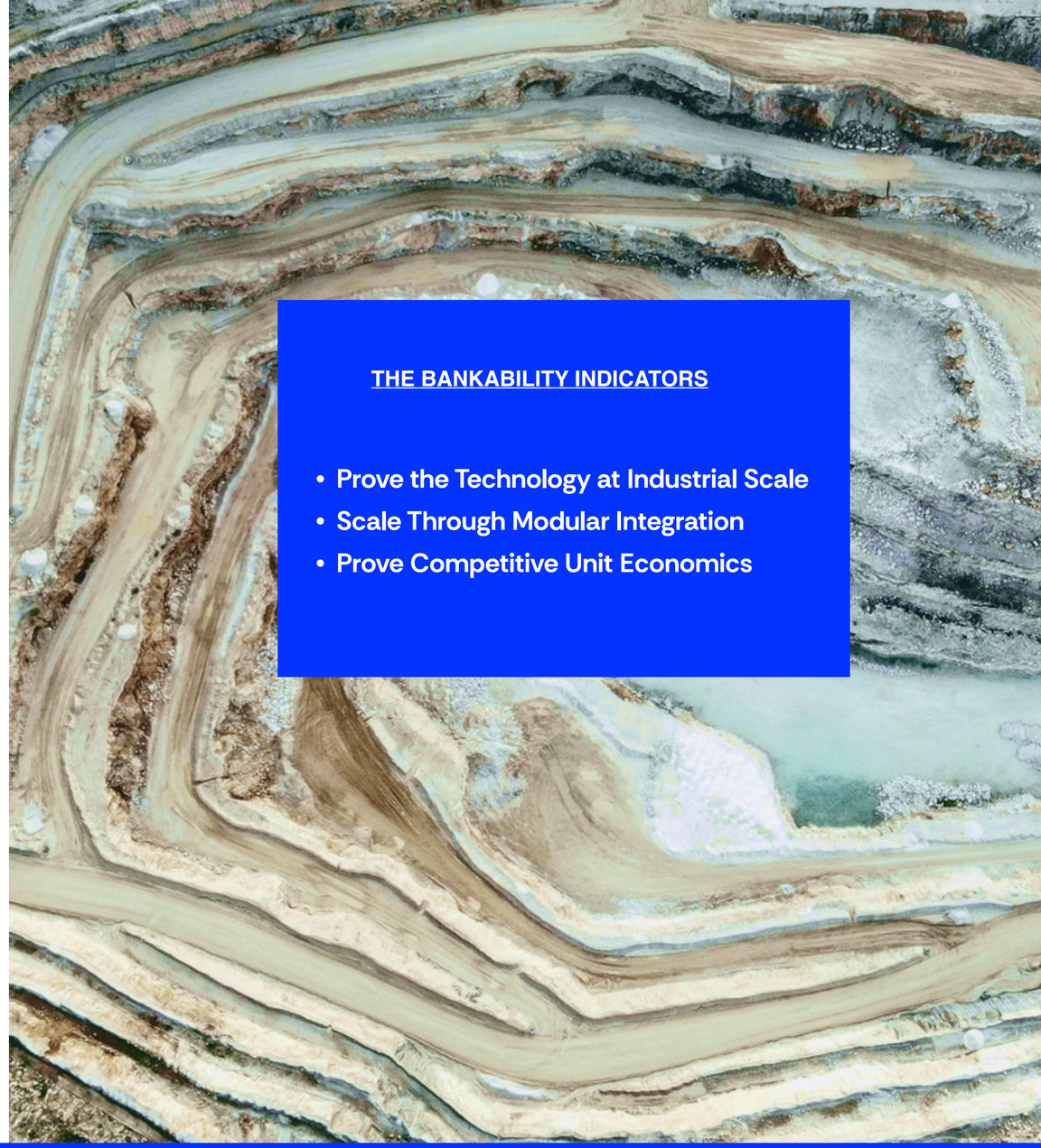
03 LOOK FOR UNIT ECONOMICS

Reliable feedstock supply, customers for recovered products, operating costs and further financing will determine whether the technology can become a scalable business.

THE INVESTOR VIEW

"In the commodity world, the cheapest, most straightforward, and most scalable technology wins."

Ananda Impact Ventures



THE BANKABILITY INDICATORS

- Prove the Technology at Industrial Scale
- Scale Through Modular Integration
- Prove Competitive Unit Economics

THE COMMERCIAL THESIS

Integrating repeatable refining modules into existing facilities could reduce capital requirements, but customer demand, cost per tonne and commercial margins remain to be validated.

from BELIEF *to* BANKABILITY

A Nordic initiative turning circular ideas into investable business

NORDIC CIRCULAR INVESTMENTS